

# Deep Limit Order Book Forecasting

A Microstructure-First Introduction

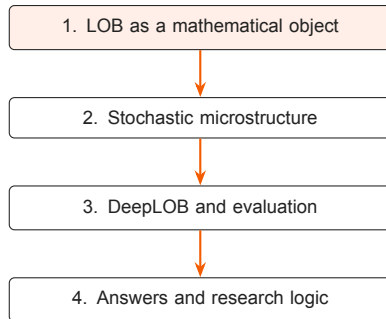
Yida Ding    May 26, 2026

## ■ Opening Questions

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### Keep these in mind

- ▶ What exactly is a LOB forecasting model predicting?
- ▶ Why do tick size, spread, and horizon change the difficulty?
- ▶ Why can a classifier be accurate but still be weak as a trading signal?



## ■ Papers and Roles

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| Paper                      | Role                                                                                             | How we use it                                                          |
|----------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Microstructural Guide      | Main paper: explains LOB predictability through market structure                                 | Labels, tick regimes, MCC, and $p_T$                                   |
| DeepLOB                    | Canonical baseline: treats the LOB as a structured time series                                   | Model principle, not engineering details                               |
| Ait-Sahalia et al.<br>HLOB | Shows short-lived high-frequency predictability<br>Encodes structural dependence inside the book | Why horizon and latency matter<br>Example of a stronger inductive bias |
| Benchmark Study            | Shows single-dataset scores are fragile                                                          | Generalization warning                                                 |

## LOB 是什么：一个状态快照

- ▶ LOB 记录还没有成交的买单和卖单。
- ▶ 每一档有价格和数量：卖一、卖二、买一、买二等。
- ▶ 深度学习常用前  $L = 10$  档，把一个快照写成向量。

### 一条 LOB 记录

$$L(\tau) = \{p_\ell^{ask}, v_\ell^{ask}, p_\ell^{bid}, v_\ell^{bid}\}_{\ell=1}^L \in \mathbb{R}^{4L}.$$

### 模型看到的输入

$$X_\tau = (L(\tau - T + 1), \dots, L(\tau))$$

$$X_\tau \in \mathbb{R}^{T \times 4L}$$

### 直觉

不是一根价格线，而是一段“买卖盘形状”的电影。

# ■ Microstructure Model: A Jump Process

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- ▶ A useful abstraction is a continuous-time state process:

$$Q_t = (Q_t^{b,1}, \dots, Q_t^{b,L}, Q_t^{a,1}, \dots, Q_t^{a,L}).$$

- ▶ Each coordinate is queue size at one bid/ask level.
- ▶ Limit orders, market orders, and cancellations are jumps of  $Q_t$ .

## Point-process view

$$dQ_t = \sum_{e \in \mathcal{E}} \nu_e(Q_{t-}) dN_t^e$$

$$\mathbb{P}(dN_t^e = 1 \mid \mathcal{F}_{t-}) \approx \lambda_e(Q_{t-}) dt$$

## Intuition

The book is not a smooth curve; it is a queueing system hit by random order-flow events.

## ■ Why This Model Explains Predictability

### Price moves when a queue is depleted

$$\tau_a = \inf\{t : Q_t^{a,1} = 0\}$$

$$\tau_b = \inf\{t : Q_t^{b,1} = 0\}$$

$$\text{next move} = \begin{cases} \text{up,} & \tau_a < \tau_b, \\ \text{down,} & \tau_b < \tau_a. \end{cases}$$

- ▶ Queue imbalance is predictive because it changes depletion probabilities.

$$I_t = \frac{Q_t^{b,1} - Q_t^{a,1}}{Q_t^{b,1} + Q_t^{a,1}}.$$

- ▶ Event clustering matters: order arrivals are often self- and cross-exciting.

$$\lambda_i(t) = \mu_i + \sum_j \int_0^t \phi_{ij}(t-s) dN_j(s).$$

- ▶ Deep models are useful because they learn nonlinear approximations to these conditional probabilities.

## ■ 预测目标：未来中间价方向

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### 中间价与价差

$$m_\tau = \frac{p_1^{ask}(\tau) + p_1^{bid}(\tau)}{2}$$
$$\sigma_\tau = p_1^{ask}(\tau) - p_1^{bid}(\tau)$$

- ▶  $m_\tau$ : 常用来定义方向。
- ▶  $\sigma_\tau$ : 买卖价差，也是交易摩擦。

### 三分类标签

$$y_\tau = \begin{cases} -1, & m_{\tau+\Delta\tau} - m_\tau \leq -\theta, \\ 0, & |m_{\tau+\Delta\tau} - m_\tau| < \theta, \\ +1, & m_{\tau+\Delta\tau} - m_\tau \geq +\theta. \end{cases}$$

- ▶  $\theta$ : tick size, 最小价格变动单位。
- ▶  $\Delta\tau$ : 未来多少次 LOB 更新，而不是固定秒数。

## ■ 主论文到底研究什么

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### 实验设置

- ▶ 15 只 NASDAQ 股票，2017–2019 年。
- ▶ LOBSTER 逐笔订单簿数据。
- ▶ 输入：最近 100 次 LOB 更新，前 10 档。
- ▶ 输出： $H_{10}, H_{50}, H_{100}$  后的方向。

### 真正的问题

- ▶ 不只是“DeepLOB 分数多少”。
- ▶ 而是：哪些股票更容易预测？为什么？
- ▶ 分类指标好，是否意味着信号可以形成完整交易？





# LOBFrame: 把流程标准化

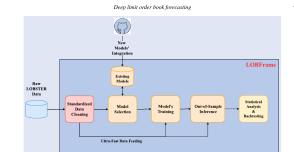


Figure 2. Pipeline representation of LOBFrame. This framework facilitates the Limit Order Book (LOB) forecasting practice through a workflow pipeline that includes data cleaning, standardized preprocessing, model selection, and integration of both existing and novel forecasting models. The LSTM layer, on the other side, is used to capture residual time dependencies among the resulting features. It is worth underlining that short time dependencies are already captured by the convolutional layers, which take LOB samples as inputs (see Figure 1). To train, validate and test the DeepLOB model, we design a high-performance data loader, which samples mini-batches of size 32 (as per the original model's implementation), each made of inputs with size  $300 \times 40$ . Dimension 300 (i.e. the temporal dimension) represents the history length and corresponds to the number of historically consecutive LOB updates constituting each sample. Dimension 40, instead, is the number of spatial constraints for each LOB's samples (see equation (1)). The sampling process differs for the training, validation, and test sets. During training, the (sub)-sampling is random and balanced. From each trading day, we draw the number of samples for the less represented class and (i) if this value is  $\geq 5000$ , then we sample 5000 random representatives (a representative is a  $100 \times 40$  input) for each of the three classes (see equation (21)), otherwise, (ii) if this value is  $< 5000$ , then we sample a number of random representatives for each class which is equal to the number of samples for the less represented class.

During validation and test stages, we still sample batches with a size of 32, but they are always sequential and cover the totality of data in the two sets. In line with the related literature (Zhang et al., 2019), the model is always trained for a maximum number of epochs equal to 100, with patience equal to 15 epochs. We use a modified version of the Adam optimizer (Kingma and Ba 2014) with decoupled weight decay (Loshchilov and Hutter 2017), commonly known as "AdamW". Following the latest applied research findings

(Bertoni et al., 2020; Karpathy, 2020), we use a learning rate equal to  $6 \times 10^{-4}$ , a  $\beta_1$  decay rate equal to 0.9 and a  $\beta_2$  decay rate equal to 0.99. The choices of values for these parameters are determined by the training pipeline described above, which is different from the one proposed in the original work (Zhang et al., 2019) and relies on a smaller number of training samples to reduce the model's exposure to the noise characterizing the LOB. The entire framework described in this paper is coded in Python using the PyTorch deep learning library (Paszke et al., 2019). A total number of 135 experiments have been run on the University College London Computer Science Department's High-Performance Computing Cluster (UCL, 2024) for a cumulative GPU runtime of 959 hours, 16 minutes and 21 seconds. Six different types of GPUs have been used: (i) NVIDIA GeForce GTX 1080 Ti; (ii) NVIDIA GeForce RTX 3080 Ti; (iii) NVIDIA TITAN X (Pascal); (iv) NVIDIA TITAN Xp; (v) Tesla V100 PCIe-16GB; and (vi) Tesla V100 PCIe-32GB.

## 6. Microstructural prices

In this section, we investigate stocks' foundational microstructural properties, focusing on the interplay between tick size and bid-ask spread, liquidity at the best levels, and the structural organization of the Limit Order Book (LOB). Through this analysis:

- We identify a practical classification standard for tick-based stock categories, identifying unique behavioral patterns across small-, medium-, and large-tick stocks and revealing the macro-stability of spread distributions over the 3-year analysis period.

Source: Briola et al., "Deep Limit Order Book Forecasting: A Microstructural Guide", Figure 2.

## ■ Tick size: 这篇文章的核心变量

### 按价差相对 tick 分类

$$\bar{\sigma} = \mathbb{E}[\sigma_{\tau}], \quad r = \frac{\bar{\sigma}}{\theta}$$

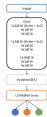
|             |                                       |
|-------------|---------------------------------------|
| small-tick  | $\bar{\sigma} > 3\theta,$             |
| medium-tick | $1.5\theta < \bar{\sigma} < 3\theta,$ |
| large-tick  | $\bar{\sigma} < 1.5\theta.$           |

- ▶ large-tick: 价差常接近 1 tick, 最优档队列很重要。
- ▶ small-tick: 价差可以展开成很多 tick, 结构更稀疏、更不稳定。
- ▶ 因此, 同样的标签规则, 在不同股票上不是同一个难度。

### 初学者理解

tick size 不只是交易所规则; 它决定“价格动一下”在这个市场里有多大意义。

## ■ DeepLOB: 模型只讲思路



signal processing [54]. FIR filters are popular smoothing techniques for denoising target signals and they are simple to implement and work well. We can write any FIR filter in the following form:

of signal  $y[n]$  at any time is a weighted sum of past values of the input signal  $x[n]$ . The constant  $a$  and  $b_k$  is the filter coefficients of the neural network, the coefficients of its are not obtained via a statistical objective from the filtration theory, but are left as degrees of freedom of the network, in order as to enhance its value

$$\{p_1^{(1)}(t), p_1^{(2)}(t), p_1^{(3)}(t), p_1^{(4)}(t), p_1^{(5)}(t), p_1^{(6)}(t), p_1^{(7)}(t), p_1^{(8)}(t), p_1^{(9)}(t), p_1^{(10)}(t), p_1^{(11)}(t), p_1^{(12)}(t), p_1^{(13)}(t), p_1^{(14)}(t), p_1^{(15)}(t), p_1^{(16)}(t), p_1^{(17)}(t), p_1^{(18)}(t), p_1^{(19)}(t), p_1^{(20)}(t), p_1^{(21)}(t), p_1^{(22)}(t), p_1^{(23)}(t), p_1^{(24)}(t), p_1^{(25)}(t), p_1^{(26)}(t), p_1^{(27)}(t), p_1^{(28)}(t), p_1^{(29)}(t), p_1^{(30)}(t), p_1^{(31)}(t), p_1^{(32)}(t), p_1^{(33)}(t), p_1^{(34)}(t), p_1^{(35)}(t), p_1^{(36)}(t), p_1^{(37)}(t), p_1^{(38)}(t), p_1^{(39)}(t), p_1^{(40)}(t)\} \quad (16)$$

$$[p_n^{(1)}(t), \sigma_n^{(1)}(t), p_n^{(2)}(t), \sigma_n^{(2)}(t)]_{n=1}^{\infty} \quad (9)$$

where  $i$  denotes the  $i$ -th level of a limit order book. The size of our first convolutional filter is  $[1 \times 2]$  with stride  $[1 \times 2]$ . The first layer essentially summarizes information between price and volume  $\{p^{(i)}, v^{(i)}\}$  at each order book level. The usage of stride is necessary here as an important property of convolutional layers is parameter sharing. This property is attractive as few parameters are estimated, largely avoiding overfitting problems. However, without strides, we would apply same parameters to  $\{p^{(i)}, v^{(i)}\}$  and  $\{p^{(i+1)}, v^{(i+1)}\}$ . In other words,  $p^{(i)}$  and  $v^{(i)}$  would share same parameters

Because the first layer only captures information at each order book level, we would expect representative features to be extracted when integrating information across multiple order book levels. We can do this by utilizing another convolutional layer with filter size  $[1 \times 2]$  and stride  $[1 \times 2]$ . The resulting feature maps actually form the micro-price defined by [53].

$$f = \frac{y_k^{(1)}}{y_k^{(1)} + y_k^{(0)}} \quad (7)$$

The weight  $l$  is called the imbalance. The micro-price is an important parameter that considers solution of  $8d$  and ask size, and the imbalance between bid and ask size is a very strong indicator of the next price movement. This feature has been reported by a variety of researchers [26–27, 58, 59, 60]. Unlike the micro-price which only the first order level is considered, we utilize correlations to three micro-prices for all levels of a LOB on the scaling frames maps. Our maps are of size  $[100, 10]$  after two layers with strides. Finally, we integrate all information by using a large filter of  $(1 \times 100)$  and the dimension of our feature maps before the Inception Module is  $(100, 1)$ .

We apply zero padding to every convolutional layer so the time dimension of our inputs does not change and Leaky Rectifying Linear Units (Leaky-ReLU) [61] are used as activation functions. The hyper-parameter (the small gradient when the unit is not active) of the Leaky-ReLU is set to 0.01, evaluated by grid search on the validation set.

Another important property of convolution is that of equivariance to translation [85]. Specifically, a function  $f(x)$  is equivalent to a function  $g$  if  $f(x) = g(f(x))$ . For example, suppose that there exists a main classification feature  $s$  located at  $(x_m, y_m)$  of an image  $I(x, y)$ . If we shift every pixel of  $I$  one unit to the right, we get a new image  $I'$  where  $I'(x, y) = I(x - 1, y)$ . We can still obtain the main

classification is done by  $\text{tr}(\mathbf{Y}^T \mathbf{Y})$  and  $m = m'$ , while the location of  $\text{tr}(\mathbf{Y}^T)$  will be at  $(x_{m'}, y_{m'}) = (x_m, -1, y_m)$ . This is important to time-series data, because convolution can find universal features that are decisive to final outputs. In our case, suppose a feature that studies imbalance is obtained at time  $t$ . If the same event happens later at time  $t'$  in the input, the exact feature can be extracted later at  $t'$ .

$$[p_n^{(1)}(t), \sigma_n^{(1)}(t), p_n^{(2)}(t), \sigma_n^{(2)}(t)]_{n=1}^{\infty} \quad (9)$$

where  $i$  denotes the  $i$ -th level of a limit order book. The size of our first convolutional filter is  $[1 \times 2]$  with stride  $[1 \times 2]$ . The first layer essentially summarizes information between price and volume  $\{p^{(i)}, v^{(i)}\}$  at each order book level. The usage of stride is necessary here as an important property of convolutional layers is parameter sharing. This property is attractive as few parameters are estimated, largely avoiding overfitting problems. However, without strides, we would apply same parameters to  $\{p^{(i)}, v^{(i)}\}$  and  $\{p^{(i+1)}, v^{(i+1)}\}$ . In other words,  $p^{(i)}$  and  $v^{(i)}$  would share same parameters

- ▶ 输入是  $100 \times 40$ : 100 次更新、10 档买卖盘价格和数量。
- ▶ CNN: 学习局部队列形状, 例如 bid/ask imbalance 和 depth slope。
- ▶ Inception: 同时看不同尺度的短期 order-flow pattern。
- ▶ LSTM: 近似条件分布

$$\mathbb{P}(y_\tau = c \mid X_\tau).$$

# 一句话

DeepLOB 好用，是因为它把随机队列过程的局部形状和短期路径依赖放进同一个函数近似器。

Source: Zhang, Zohren and Roberts.

“Deep Learning for Limit Order Books”, Figure 3.

## ■ 指标 1: MCC 比准确率更稳

- ▶ LOB 标签常常不均衡，很多时候“稳定”类占比高。
- ▶ accuracy 可能被多数类抬高。
- ▶ MCC 可以理解为“预测标签和真实标签的相关性”。
- ▶ 随机预测约为 0，完美预测为 1。

### 混淆矩阵视角

$MCC = F(\text{全部 true/false positive/negative})$

它不只看“猜对多少”，还看各类之间怎么错。

### 主论文发现

large-tick 股票在多个 horizon 上更稳定、更容易预测；这说明 score 反映的是模型与市场结构的共同结果。

# 主论文关键图：tick 类别改变结果

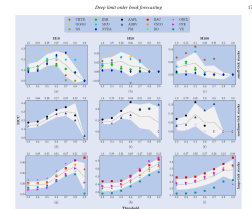


Figure 10. Average Modified Correlation Coefficient (MCC). Results are separated according to the prediction horizons taken into account (one, three, and twelve tick size) and the corresponding standard deviation (shown through the grey line and shadow). All the average values and the standard deviation are computed by considering stocks with the same tick size, ignoring the 1-year analysis period.

has a value of 0.078. Differently from what happens for the other two classes of stocks, a clear average growing pattern is detected for threshold values  $\leq 0.35$ , while the average percentage of data used to compute the metric decreases more than what happened at H50, containing, however, higher than the minimum threshold for small- and medium-tick stocks. Overall, large-tick stocks demonstrate greater forecast stability across horizons, as evidenced by the lower variance in MCC. This robustness is strongly linked to their higher liquidity and less brutal spread distributions, which provide a more stable market structure for the DeepLOB model to learn from.

The analysis reported in this Section is further deepened in Appendixes 2 and 3, where we report (i) the one-star MCC of the DeepLOB model at  $H_{10} \in \{10, 50, 100\}$ , for different confidence levels; (ii) the corresponding statistical significance; and (iii) a replica of the analysis in Figure 10 for the FI and accuracy score. The coherence of the results between the robustness of the findings discussed earlier in this Section, highlighting that large-tick stocks exhibit a significant predictability, one across all the considered horizons.

This is evidenced in Figures A1 and A2, where we observe that those stocks whose FI score realizations greater than 0.45 without applying any thresholds and surpass 0.7 when probability thresholds are implemented. Similarly, accuracy scores exceed 0.7 without thresholds and reach over 0.9 with the application of probability thresholds. However, as detailed in section 7.2, achieving high scores on these traditional machine learning metrics does not necessarily translate into the generation of actionable trading signals. This distinction remarks the complexity of converting predictive accuracy into practical trading strategies.

## 7.2 On the predictability of model's forecasts

The analysis presented above offers significant insights into the DeepLOB model's performance at different horizons for different classes of stocks. However, further discussion is needed to understand the results from the perspective of the microstructural properties of the LOB. To do so, in this Section, we introduce a novel methodology to evaluate the

- ▶ 行: small / medium / large tick。
- ▶ 列:  $H_{10}, H_{50}, H_{100}$ 。
- ▶ 横轴: 只保留更高置信度预测。
- ▶ 读图结论: 模型表现不是单纯由网络决定, 而是和股票微观结构强相关。

Source: Briola et al., Microstructural Guide,

Figure 10.

## ■ 指标 2：为什么还需要 $p_T$

- ▶ F1/MCC 是逐样本指标：每个时间点算一次对错。
- ▶ 交易是路径问题：需要开仓、持仓、平仓都大致对。
- ▶ 一个错误如果刚好发生在平仓点，影响比普通点更大。

### 完整交易概率

$P_T = \{\text{模型预测出的完整交易}\}$

$T_T = \{\text{真实方向对应的完整交易}\}$

$$p_T = \frac{|P_T \cap T_T|}{|P_T \cup T_T|}$$

### 直觉

$p_T$  关心“预测序列能否形成正确交易”，而不只是“单点标签是否正确”。

## ■ 主论文结论：用一句话讲通

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1. LOB 可预测性不是模型单独决定的，而是由模型、标签、horizon、tick regime 共同决定。
2. large-tick 股票通常更稳定：价差窄、最优档流动性厚，队列变化更有信息。
3. 传统分类指标有用，但不够；交易信号还要看时间顺序和完整交易路径。
4. 这篇文章真正的贡献是：把深度学习预测重新放回市场微观结构里解释。

### 适合初学者的记法

先理解市场结构，再评价模型分数；不要反过来只看 leaderboard。

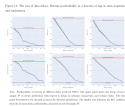
## ■ 其它论文速览（一）：信号与模型

### Ait-Sahalia et al.

- ▶ 高频收益/方向确实存在短寿命可预测性。
- ▶ 延迟会快速消耗预测价值。
- ▶ 作用：解释为什么 LOB 信号必须看极短 horizon。

### DeepLOB

- ▶ 经典 CNN + Inception + LSTM 结构。
- ▶ 作用：提供主论文使用的标准模型基线。
- ▶ 局限：不能单独解决跨市场和交易可用性问题。



Source: Ait-Sahalia, Kalnina and Xiu, Figure 14.





## ■ 五篇论文合成一个框架

| 层次      | 要问的问题                            | 对应论文                   |
|---------|----------------------------------|------------------------|
| 信号是否存在  | 极短 horizon 上是否有可预测性？延迟伤害多大？      | Ait-Sahalia et al.     |
| 输入怎么表示  | LOB 是价格序列，还是买卖盘形状序列？             | DeepLOB                |
| 市场结构是什么 | tick size、spread、depth 是否改变预测难度？ | Microstructural Guide  |
| 分数是否可信  | MCC 高是否代表完整交易可用？                 | Microstructural Guide  |
| 模型能否泛化  | 换股票、换市场、换 horizon 后排序是否稳定？       | HLOB / Benchmark Study |

## ■ Principle-First Research Logic

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1. Define the random object: state  $Q_t$ , observation  $X_\tau$ , and label  $y_\tau$ .
2. Explain the market regime:  $\bar{\sigma}/\theta$ , best-queue depth, imbalance, event intensity.
3. Choose a model bias that matches the object: local convolution, recurrence, graph structure, or point-process intensity.
4. Evaluate two levels: sample-wise correctness and path-wise tradability.
5. Test whether the explanation survives across stocks, horizons, and market regimes.

### 原则

Model complexity should serve a microstructure explanation; it should not replace one.

## ■ Answers to the Opening Questions

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1. What is being predicted?

The future direction of the mid-price, usually after a fixed number of LOB updates:

$$y_\tau = g_\theta(m_{\tau+\Delta\tau} - m_\tau).$$

2. Why do tick size, spread, and horizon matter?

They change queue depletion probabilities, class balance, signal lifetime, and transaction-cost meaning.

3. Why is high accuracy not enough?

A trading signal is a time-ordered path. MCC/F1 count point errors, while  $p_T$  asks whether the predicted path completes the right trade.

## ■ 最后的 takeaway

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1. LOB 预测的核心对象是“订单簿状态序列”，不是单纯价格序列。
2. LOB 可以看成由订单到达、成交和撤单驱动的随机队列过程。
3. tick size 是理解可预测性的入口变量，因为它改变队列、价差和标签的尺度。
4. DeepLOB 是一个好 baseline，但主论文强调的是何时有效、为什么有效。
5. MCC 比准确率更适合不均衡分类； $p_T$  更贴近交易路径。
6. 真实研究不能只看一个 benchmark 分数。

### 一句收束

不要先问“哪个模型最强”；先问“这个市场结构下，什么方向信号在什么 horizon 上可预测，并且能否形成完整交易路径”。

## ■ 讨论问题

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- ▶ 如果 MCC 上升但  $p_T$  下降，这个模型该怎么用？
- ▶ 对 small-tick 股票，应优先改标签、改输入表示，还是改模型结构？
- ▶ 事件时间 horizon 和物理时间 horizon，哪个更适合我们的策略目标？

Q & A

## ■ 文献与图表来源

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- ▶ A. Briola, S. Bartolucci, and T. Aste, “Deep Limit Order Book Forecasting: A Microstructural Guide,” Quantitative Finance, 2025.
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- ▶ E. Bacry, I. Mastromatteo, and J.-F. Muzy, “Hawkes Processes in Finance,” Market Microstructure and Liquidity, 2015.
- ▶ A. Briola et al., “Information Persistence and Structure in Limit Order Books.”
- ▶ “LOB-Based Deep Learning Models: Benchmark Study.”

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